PRODUCT GUIDE

Lumora Room Sensor S1 Product Guide

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Purpose. Reference for supporting the Lumora Room Sensor S1 (model LS-S1), a battery powered temperature and occupancy sensor used with the Thermostat T2, Thermostat T3 and Hub H1.

1. Product Overview

The Room Sensor S1 measures temperature and detects occupancy in the room where it is placed. Thermostats use these readings to balance comfort between rooms. The S1 communicates over the LumoraLink low-power radio and does not connect to the home wireless network directly.

The S1 is sold as a single unit (SKU LS-S1-01, $39.00) and as a two-pack (SKU LS-S1-02, $69.00). It can be paired with a Thermostat T2 (maximum 2 sensors), a Thermostat T3 (maximum 6 sensors) or a Hub H1, which allows the sensor to be used for automations without a thermostat.

2. Specifications

Item	Room Sensor S1 (LS-S1)
Sensors	Temperature, passive infrared (PIR) occupancy
Temperature accuracy	±0.9 °F (±0.5 °C)
Occupancy detection	Up to 15 ft (4.5 m), 100° horizontal field of view
Radio	LumoraLink 915 MHz
Range	Up to 60 ft (18 m) indoors to the thermostat or hub
Battery	One CR2477 3 V lithium coin cell (included)
Battery life	Up to 2 years at the default 5-minute reporting interval
Low battery warning	Sent to the app at 15% remaining
Operating environment	32 °F to 104 °F, indoor use only
Dimensions	1.8 x 1.8 x 0.6 in (46 x 46 x 15 mm)
Warranty	1 year (12 months) limited warranty

Table 2-1. Room Sensor S1 specifications.

Setting the reporting interval to 1 minute in the app (Sensors > Advanced) improves responsiveness but reduces battery life to about 8 months. The interval cannot be set longer than 10 minutes.

3. Placement and Pairing

3.1 Placement Guidelines

- Place the sensor about 5 ft (1.5 m) above the floor on an interior wall or shelf.
- Keep it at least 3 ft away from windows, exterior doors, supply vents, fireplaces and electronics that produce heat.
- For occupancy detection, point the lens toward the main seating or sleeping area. The sensor does not detect motion through glass.
- Do not install the S1 in bathrooms, garages, attics or outdoors.

3.2 Pairing Procedure

1. In the Lumora Home app, open the thermostat or hub and tap Add Sensor.
2. Pull the battery tab out of the sensor. The LED blinks blue to show it is in pairing mode for 3 minutes.
3. Hold the sensor within 30 ft (9 m) of the thermostat or hub during pairing; it can be moved to its final location afterwards.
4. Wait for three green blinks, then name the room in the app.
5. To re-enter pairing mode later, press and hold the button on the back for 5 seconds until the LED blinks blue.

4. LED Indicators

LED pattern	Meaning	Action
Blinking blue	Pairing mode (3 minutes)	Complete pairing in the app.
Three green blinks	Paired successfully	None.
Single green blink	Occupancy detected (first 30 minutes after pairing only)	None; used for placement testing.
Red double blink	Pairing failed	Move closer and retry pairing.
Slow amber blink every 60 s	Battery below 15%	Replace the CR2477 battery.
Solid red for 5 s	Hardware fault	Remove battery for 30 s; if repeated, warranty claim.

Table 4-1. S1 LED patterns.

5. Troubleshooting

5.1 Sensor Shows Offline

1. Check the battery level in the app. Replace the battery if it is below 15% or if the sensor has been offline for more than 2 hours (thermostats report error E5).
2. Confirm the sensor is within 60 ft of the thermostat or hub. Metal appliances, mirrors and concrete walls reduce range significantly.
3. Press the button once. If the LED does not light at all, the battery is fully discharged or inserted upside down (the + side faces the cover).
4. Remove the sensor from the app and pair it again.

5.2 Temperature Differs From the Thermostat

Differences of up to 2 °F between rooms are normal. If a sensor reads more than 2 °F differently from a reference thermometer placed beside it for 30 minutes, apply an offset in the app (up to ±5 °F). A difference above 5 °F after an offset indicates a faulty sensor and qualifies for a warranty claim.

5.3 Occupancy Not Detected

PIR sensors detect movement of warm bodies. A person sitting still may not be detected for several minutes. The thermostat keeps a room marked as occupied for 30 minutes after the last detection. Pets under 40 lb are usually ignored when the sensor is mounted at 5 ft.

6. Battery Replacement

1. Twist the back cover counter-clockwise and lift it off.
2. Replace the CR2477 cell with the + side facing the cover. A CR2032 will not fit and must not be used.

3. Close the cover. The sensor reconnects automatically within 2 minutes; no re-pairing is needed.

> **Note:** Batteries are consumables and are not covered by the S1 warranty. Customer Care may send one free replacement battery per sensor during the first year as a goodwill gesture.

7. Comfort Strategies With Several Sensors

Customers often ask how the thermostat uses several sensors. Agents can explain the three comfort modes available in the app under Comfort > Sensors:

Mode	How it works	Best for
Average	Averages the thermostat and all selected sensors equally	Open-plan homes with small differences between rooms
Priority room	Uses only one selected room for the current schedule period	Bedrooms at night, home office during the day
Follow Me (T3 only)	Averages only rooms with occupancy in the last 30 minutes	Homes where people move between rooms

Table 7-1. Sensor comfort modes.

The T2 supports the Average and Priority room modes only. With either thermostat, a sensor that has not reported for 2 hours is removed from the calculation until it reports again.

8. Quick Answers for Agents

Can the S1 be used outdoors or in a garage?

No. The S1 is rated for indoor use from 32 °F to 104 °F. Readings outside this range are not reliable and condensation can damage the sensor.

Does the S1 measure humidity?

No. The S1 measures temperature and occupancy only. Humidity is measured by the Thermostat T3.

Can one sensor be shared by two thermostats?

No. Each sensor is paired with exactly one thermostat or hub. In homes with two zones, pair each sensor with the thermostat of the zone where it is installed.

Is the S1 a camera or microphone?

No. The PIR sensor only detects changes in infrared heat. It cannot record images or sound.

9. Appendix: Placement Examples by Room

Room	Recommended placement	Avoid
Bedroom	Dresser or wall facing the bed, 4-5 ft high	Nightstand next to a lamp or phone charger
Living room	Bookshelf facing the seating area	Above a TV or near a fireplace
Home office	Wall facing the desk	On the desk next to a computer
Nursery	Wall away from the crib, out of reach	Inside the crib or on the floor
Kitchen	Not recommended	Cooking heat causes false readings
Basement living area	Interior wall away from the furnace	Near the furnace, water heater or sump pump

Table 9-1. Placement examples.

The S1 is small enough to swallow and the coin cell is hazardous if swallowed. In nurseries and children's rooms, always mount the sensor out of reach using the included adhesive pad or screw mount, and confirm the back cover is fully closed.

10. Warranty

The Room Sensor S1 carries a 1-year (12-month) limited warranty. Sensors are not eligible for advance replacement unless the customer has an active Lumora Plus subscription. See the Lumora Limited Warranty Policy (LHS-POL-001) for full terms.